Samantha Freeland
Relational Algebra

Q1 ← π AnimalID, ANIMAL.Name, HabitatID, HABITAT.Name, HABITAT.Type (σ Zoo_ID = 20 (ANIMAL ⋈ Zoo_ID = Zoo_ID HABITAT))

Q2 ← π AnimalID, ANIMAL.Name, Food, Serving, ZOO.City, ZOO.Country (σ SPECIES.Genus = 'Leontopithecus', 'Mandrillus', 'Cebus', 'Alouatta', 'Ateles', 'Nasalis', 'Macaca', 'Saimiri', 'Theropithecus', 'Pan', 'Pongo' and IronLevels = 'Iron Deficiency' and Country = 'United States' (ANIMAL ⋈ Species_ID = SpeciesID SPECIES ⋈ Animal_ID = AnimalID MEDICAL ⋈ Diet_ID = DietID DIET ⋈ Zoo_ID = ZooID ZOO))

Q3 ← π AnimalID, SPECIES.Name, SPECIES.ExtinctionProbaility, SPECIES.ExpectedExtinctionDate (σ Year_Assessed = 2000 and ExtinctionProbability > 0.8 (ANIMAL ⋈ Species_ID = SpeciesID SPECIES ⋈ SpeciesID = Species_ID ENDANGERMENT))

Q4 ← π AnimalID, ANIMAL.Name, SPECIES.Name, ZOO.Name, ZOO.City, ZOO.Country, HABITAT.Name, HABITAT.Type (ANIMAL ⋈ Habitat_ID = HabitatID HABITAT ⋈ Species_ID = SpeciesID SPECIES)

Q5 ← 𝓕 Continent, SPECIES.Name, AVG(ANIMAL.Weight) (σ SPECIES.Genus = 'Panthera' ((ANIMAL ⋈ Species_ID = SpeciesID) ⋈ Zoo_ID = ZooID ZOO))

Q6 ← π SPECIES.Name, ANIMAL.Gender, COUNT(ANIMAL.AnimalID), ENDANGERMENT.Extinction.Probability (𝓕 SPECIES.Name, ANIMAL.Gender, ENDANGERMENT..ExtinctionProbability, COUNT(ANIMAL.AnimalID) (σ ExtinctionProbability > 0.5 ((ANIMAL ⋈ Species_ID = SpeciesID SPECIES) ⋈ ENDANGERMENT..Year_Assessed = 2023 and ENDANGERMENT.ExtinctionProbability > 0.5)))